Week 1 : 6CS030 Big Data

Python Revision

In this lab exercise, we will learn the basics of python programming language. This basics will aid us in the future lab sessions.

> First of all, you need to download this python interpreter. To write the python script in your local host, you can download various IDE's such as VSCode, Atom, PyCharm etc. However, I would suggest you to use Jupyter Notebook. Check out this video (From 4:30 seconds) to learn how to use Jupyter Notebook more efficiently.

Finally, at the end of the lab, there are few exercises to do by your own.

Let's Begin 😀

Python Fundamentals

Python is an **interpreted** programming language. It can be used to develop programs based on procedural and object-oriented programming paradigms. Python has a very simple syntax. It provides many built-in programming features. Primitive Types There are several primitive types in python. Primitive type means a built-in type which is a part of the language. In short, python has some built-in data types and data structures which ease the overall coding experience. The most important ones are the followings:

> 1. Integer (int)
> 2. Floating Point (float)
> 3. Boolean Logic (bool)
> 4. String (str)
> 5. List ('list' or can be denoted as square brackets)
> 6. Tuple ('tuple' or can be denoted as paranthesis)
> 7. Dictionary ('dict' or can be denoted as curly braces)

Integer: numeric values without fractional part. Examples: 1, 32, 999

Floating Point: numeric values with fractional part. Examples: 1.3, 3.2, 8.3454

boolean: logical values. Either true or false.

string: an ordered sequence of alphanumeric characters. Examples: 'Wolves', 'Python'
list: an ordered sequence of objects in which each element is identifed by an integer subscript. Examples: [], [1, 3, 4], ['a', 3, 7.5]

tuple: similar to the list type, but the content cannot be changed once it is created. Examples: (), (1, 3, 4), ('a', 3, 7.5)

dictionary: a collection of objects in which each object contains a key and a value. Examples: {}, {1: 'orange', 2: 'banana'}

Standard Input and Output

```
"""To ask a value from the end user and store it to reuse later, fir
The "input" keyword helps to take value from the end user """
first_name = input("Please Enter Your first Name: ")
```

```
"""To show the output python has a keyword "print" which prints the
print(first_name)
```

```
rohit
```

Control Structure

> It is used to change the flow of control and execute a statement or a block of statements if some condition holds

```
"""Here we are asking user to enter a value and checks if its negati
by the control structure"""
value = int(input("Please Enter a Number: ")) #input keyword always
#it is called casting. It converts a value to different data types
if value < 0:
 print('The number is negative')
elif value == 0:
 print('The number is non-negative')
else:
 print('The number is positive')

```

```
The number is positive
```

Loops

> There are few types of iterative processes which we call loops. but the most used ones are for and while loop depending the situation

```
"""Find the sum of all integers from 1 to 10 using while loop"""
total = 0
counter = 1
while counter <= 10: #stopping condition
 total = total + counter #sum
 counter += 1 #increament
print("total", total)
```

```
total 55
```

How would you do it with for?

```
"""Find the sum of all integers from 1 to 10 using while loop"""
total = 0
for loop_count, num in enumerate(range(1,11,1)):
  total += num
print("total", total)
```

```
total 55
```

Or I would I just..

```
print("total", sum(range(1,11,1)))
```

```
total 55
```

Just the magic of :

List Modification

```
"""A list can be created using a comma-separated sequence within squ
first_list = [1, 2, 3, 8]

"""In a list, the individual items can be accessed by subscript/inde
print("First value of the list is ", first_list[0])
print("Last value of the list is ", first_list[-1])

```

```
First value of the list is  1
Last value of the list is  8
```

```
"""The elements of a list can also be accessed by traversing the lis
for values in first_list:
 print(values)

```

```
1
2
3
8
```

Other List Operations

```
#append
first_list.append(10)
print("After adding another element in the list, the list is ", firs
#modify
first_list[2] = 5
print("After modifying an element in the list, the list is ", first_
#insert: elements can be inserted in any positon/index
first_list.insert(2, 12)
print("After inserting an element in the list in the third position,
```

```
After adding another element in the list, the list is  [1, 2, 3, 8, 10]
After modifying an element in the list, the list is  [1, 2, 5, 8, 10]
After inserting an element in the list in the third position, the list is
```

There are also other methods for several operations such as remove, pop, delete etc. Please practice those by your own

Dictionary Modification

```
"""There are lot of ways to access elements of the dictionary. Howev
The traversal using the iterator mechanism (for statement) is made o
zipCode = {'London':290, 'Leeds':578, 'Wolverhampton':911}
for key in zipCode:
 print(key, '= ', zipCode[key])

```

```
London =  290
Leeds =  578
Wolverhampton =  911
```

```
"""To explicitely find the list of keys or list value we can use the
key_list = zipCode.keys()
value_list = zipCode.values()
print(key_list)
print(value_list)
```

```
dict_keys(['London', 'Leeds', 'Wolverhampton'])
dict_values([290, 578, 911])
```

```
"""To change the value of a corresponding key we can do the followin
zipCode['London'] = 666
for key in zipCode:
```

```
print(key, '= ', zipCode[key])

```

```
London =  666
Leeds =  578
Wolverhampton =  911
```

```
"""Using the same technique we can add another entry. However, the k
Duplicate key name will modify the value only"""
zipCode['Birmingham'] = 111
for key in zipCode:
 print(key, '= ', zipCode[key])

```

```
London =  666
Leeds =  578
Wolverhampton =  911
Birmingham =  111
```

Similarly to the list operations, dictionary has also lots of other operations and students are encouraged to practice those by their own

Functions

> Writing a function is pretty easy in python using the keyword def(). After declaring the function name, we have put the arguments. We do not have to mention explicitely if the passing argument is a list, dictionary or int etc.. It can be anything

```
def oddEven(num):
 if num%2 == 0:
  print('The number is even')
 else:
  print('The number is odd')
 return
num = int(input("Please Enter a Number: "))
oddEven(num)
```

```
The number is odd
```

Indentations are important so always use tab space

```
def oddEven(num):
  if num%2 == 0:
    print('The number is even')
  else:
    print('The number is odd')
  return
num = int(input("Please Enter a Number: "))
oddEven(num)
```

```
The number is even
```

Exercises

> Now that, you have known some basics of python, try to do the following exercises. These exercise questions are not limited to the functions that are given above. You have to think for the solutions and search for the functions that you need to use. In this way, you will learn a lot of operations of different data structures such as lists, dictionaries etc. This will aid you to be a better developer in python programming language as well as data analytic domain.

Exercise on Loops

Question 1: Display **Fibonacci series** up to 10 terms.

Question 2: Find the factorial of a given number. Ask the user to provide the number.

Question 3: Write a program to print a right angle triangle using the for loop.

```
def fibo(n):
  a = 0;
  b = 1;
  print(a)
  print(b)

  for i in range(0, n - 2):
    c = a + b;
    print(c)
    a = b;
    b = c;

fibo(10)
```

```
0
1
1
2
3
5
8
13
21
34
```

```
def factorial(n):
  fact = 1;

  for i in range(1, n + 1):
    fact = i * fact;

  return fact;

num = int(input("Enter a number"));
print(f"Factorial of {num} is {factorial(num)}")
```

```
Factorial of 3 is 6
```

```
def pattern(n):
  for i in range(1, n + 1):
    for j in range(0, i):
      print('*', end=" ")
    print()

pattern(5)
```

```
*
* *
```

```
* * *
* * * *
* * * * *
```

Exercise on Functions

Question 1: Make a simple calculator that can add, subtract, multiply and divide two numbers using function.

```
def calculator(a, b, operation):
  match operation.lower():
    case "add" | "+":
      return a + b;
    case "sub" | "-":
      return a - b;
    case "mul" | "*":
      return a * b;
    case "divide" | "/":
      if (b == 0): raise Exception("Cannot Divide by 0")
      return a * b;
    case _:
      return "Invalid Operation"

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
op = input("Enter operation: ")
calculated = calculator(num1, num2, op);
print(f"{num1} {op} {num2} = {calculated}")

```

```
2 + 3 = 5
```

Exercise on List and Dictionaries

Question 1: Remove the first duplicate element of a list

Question 2: Write a Python program to create a new dictionary by extracting the mentioned keys from the below dictionary.

> sample_dict = { "name": "Anderson", "age": 35, "salary": 5000, "city": "London"}

> keys = ["name", "salary"]

> **Expected Output: {'name': 'Anderson', 'salary': 5000}**

Question 3: Write a Python program to convert two lists into a dictionary in a way that item from list_one is the key and item from list_two is the value.

> list_one = ['one', 'two', 'three'] .list_two = [1, 2, 3]

> **Expected Output: {'one': 1, 'two': 2, 'three': 3}**

```
def removeDuplicate(l):
  return [*set(l)]

removeDuplicate([1, 2, 1, 1, 4])
```

```
[1, 2, 4]
```

```
def filter_dict(dict, keys):
  seen = {};

  for key in keys:
    for dict_key, val in dict.items():
      if key == dict_key:
        seen[dict_key] = val

  return seen;

sample_dict = { "name": "Anderson", "age": 35, "salary": 5000, "city
keys = ["name", "salary"]
filter_dict(sample_dict, keys)
```

```
{'name': 'Anderson', 'salary': 5000}
```

```
def convert_list(l1, l2):
  new_dict = {}
  for idx, key in enumerate(l1):
      new_dict[key] = l2[idx]

  return new_dict;

list_one = ['one', 'two', 'three']
list_two = [1, 2, 3]
convert_list(list_one, list_two)

```

```
{'one': 1, 'two': 2, 'three': 3}
```

Double-click (or enter) to edit